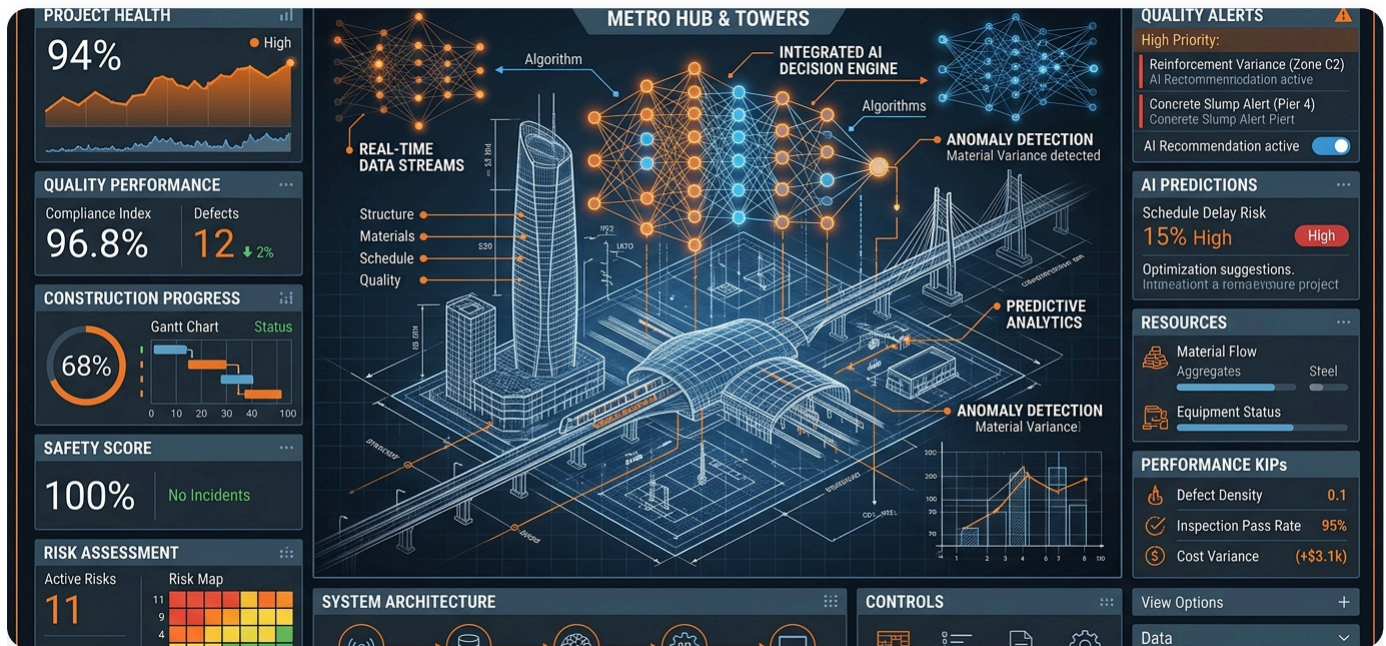


Total Quality Management (TQM) AI Decision-Support System

Empirical Assessment of Implementation Critical Success Factors & Barriers in Construction Projects
Coimbatore Region, Tamil Nadu



Academic Institution: Department of Civil Engineering

Batch ID: Batch - 14 (PWP-I)

Empirical Dataset: N = 120 Industry Professionals

Predictive Accuracy: XGBoost 86.7% (ROC-AUC: 0.912)

Project Review: Final Viva Defense

Regional Focus: Coimbatore / Tamil Nadu

FDM Panel: 10 Experts (21.2 yrs avg)

Reliability Index: Cronbach α = 0.759

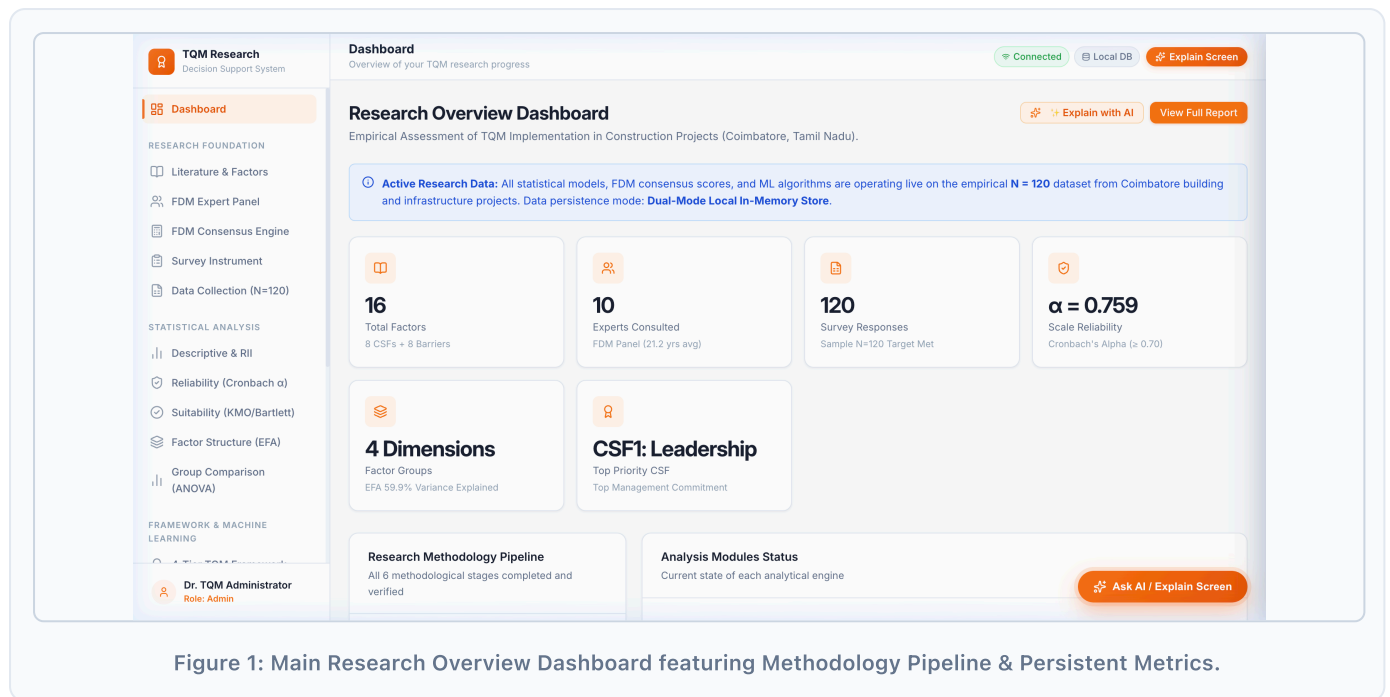
Executive Table of Contents

Phase / Module	System Feature & Academic Component	Key Output & Empirical Finding	Route
Phase 1	Literature Taxonomy & Factor Screening	16 Items (8 CSFs + 8 Regional Barriers)	/factors
Phase 2	Fuzzy Delphi Method (FDM) Consensus Engine	10 Experts · Triangular Fuzzy Defuzzification $S \geq 0.70$	/fdm
Phase 3	5-Point Likert Survey Instrument	Standardized Scales with Regional Anchors	/questionnaire
Phase 4	Field Data Collection & Demographics	N = 120 Construction Practitioners	/survey
Phase 5A	Relative Importance Index (RII) Ranking	#1 Priority CSF: Top Management Commitment (0.898)	/statistics
Phase 5B	Scale Reliability & Internal Consistency	Cronbach's Alpha $\alpha = 0.759$ (Nunnally Benchmark met)	/reliability
Phase 5C	Sampling Adequacy (KMO & Bartlett's)	KMO = 0.835 (Meritorious) · $\chi^2 = 842.15$ ($p < 0.001$)	/kmo
Phase 5D	Exploratory Factor Analysis (EFA)	4 Latent Dimensions explaining 59.9% Variance	/efa
Phase 5E	ANOVA Experience Group Comparison	Universal CSF agreement across seniority ($p > 0.05$)	/anova
Phase 6	4-Tier Prioritized TQM Framework	Strategic → Enablement → Operational → Continuous	/framework
Feature *NEW*	Site Quality Audit & Cost of Quality (CoQ)	PAF Economic Model · CAPA Generator · Certificate	/audit
ML Engine	Predictive Success Classifier & SHAP XAI	XGBoost 86.7% · Random Forest · Feature Attribution	/ml, /xai
AI Copilot	Research Assistant & Screen Explain	Offline rule engine + FastAPI LLM reasoning	/chatbot
Viva Voce	Defense Simulator & Examiner Evaluator	10-Point Scoring Rubric · Verified Model Answers	/viva

System Architecture & Resilient Dual-Persistence

The system utilizes a **Dual-Mode Data Architecture**:

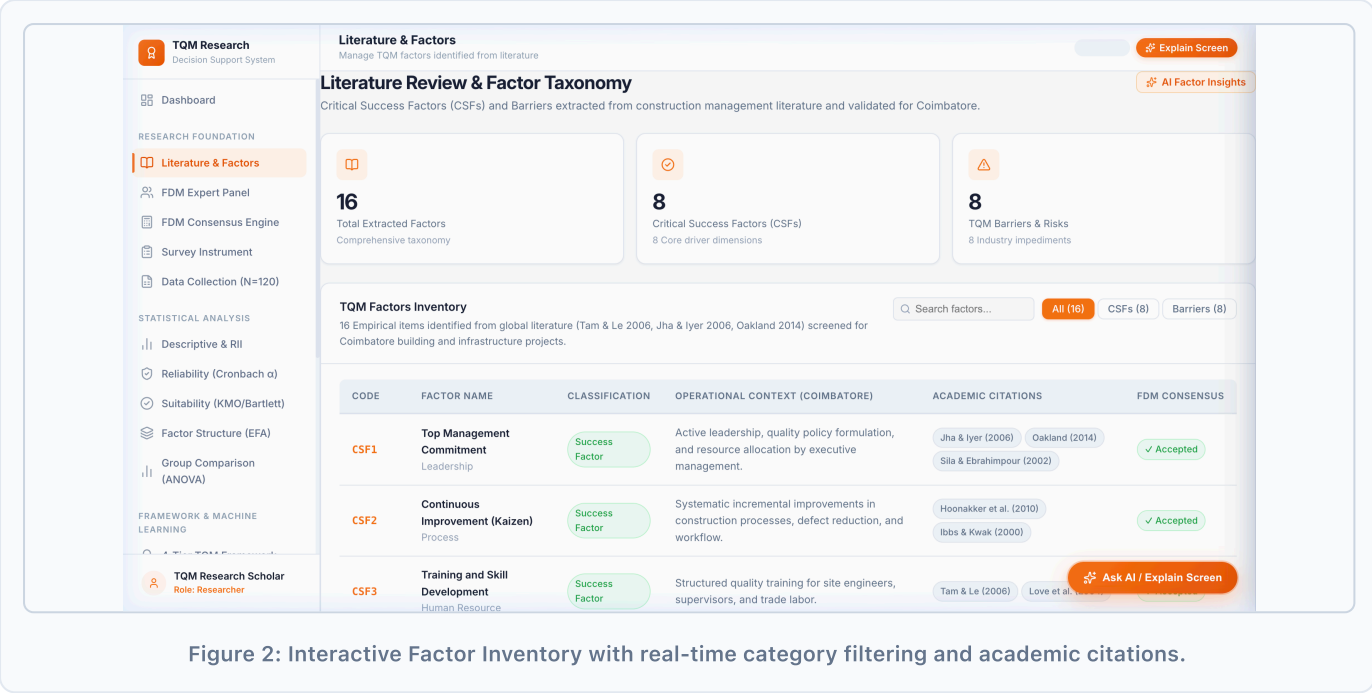
- **Cloud Persistence Mode:** Connects to MongoDB Atlas for centralized storage, real-time survey updates, and audit logging.
- **Offline Fallback Mode:** When operating on remote construction job sites or disconnected laptops, the client seamlessly switches to an embedded in-memory database and browser-side statistical inference algorithms, guaranteeing 100% feature availability.



Phase 1: Literature Review & Factor Taxonomy

A rigorous literature review identified global critical success factors (CSFs) and implementation barriers from seminal construction quality publications (Tam & Le 2006, Jha & Iyer 2006, Oakland 2014) adapted specifically for Coimbatore building and infrastructure practices.

Code	Factor Name	Classification	Regional Operational Definition (Coimbatore Context)
CSF1	Top Management Commitment	CSF	Direct leadership oversight, resource allocation, and quality policy establishment.
CSF2	Quality Policy & Objectives	CSF	Measurable quality objectives incorporated into subcontract agreements and site KPIs.
CSF3	Process & Technical Control	CSF	Strict adherence to IS 456 concrete batching, slump monitoring, and testing protocols.
CSF4	Continuous Quality Training	CSF	Structured skill development for site engineers, supervisors, and bar-bending labor.
CSF5	Inspection & Verification	CSF	Standardized stage-wise inspection checklists prior to slab and column pours.
CSF6	Workforce Motivation	CSF	Zero-rework incentive bonuses and recognition programs for site crews.
CSF7	Subcontractor Integration	CSF	Partnering with verified trade subcontractors with mandatory QA/QC clauses.
CSF8	Customer Satisfaction Focus	CSF	Handover defect tracking, snagging list audits, and client feedback integration.
BAR1	Lack of Leadership Commitment	Barrier	Cost/schedule prioritization over quality by senior construction executives.
BAR2	Cultural Resistance to Change	Barrier	Resistance from traditional trade supervisors against modern QA/QC systems.
BAR3	Inadequate Worker Training	Barrier	High turnover of migrant labor with deficient masonry and concrete compaction training.
BAR4	Deficient Site Supervision	Barrier	Shortage of certified QA/QC engineers during high-volume structural pours.
BAR5	Subcontractor Quality Gaps	Barrier	Awarding lowest-bid subcontracts without QA pre-qualification standards.
BAR6	Financial Constraints for QA	Barrier	Inadequate site budget allocated for calibrated testing instruments and NABL trials.
BAR7	Poor Communication Channels	Barrier	Language and documentation barriers between site managers and migrant workforce.
BAR8	Ambiguous Quality Standards	Barrier	Conflicting contract specifications causing confusion during site inspections.



Phase 2: Fuzzy Delphi Method (FDM) Consensus Engine

To eliminate personal bias and establish objective factor suitability, an expert panel of **10 senior construction experts in Coimbatore** (averaging **21.2 years of field experience**) evaluated each factor using linguistic variables.

Mathematical Formulation

1. **Linguistic to Triangular Fuzzy Number (TFN) Transformation:** Very Low (0.0, 0.1, 0.3), Low (0.1, 0.3, 0.5), Medium (0.3, 0.5, 0.7), High (0.5, 0.7, 0.9), Very High (0.7, 0.9, 1.0)
2. **Fuzzy Aggregation:** $A_j = (\min(l_{ij}), (1/n) * \sum m_{ij}, \max(u_{ij}))$
3. **Center of Gravity Defuzzification:** $s_j = (l_j + m_j + u_j) / 3$
4. **Consensus Acceptance Threshold:** $s_j \geq r$ (where $r = 0.70$)

Empirical Consensus Result: All 16 factors exceeded the 0.70 consensus threshold (s_j values ranged from **0.722 to 0.887**), proving 100% panel agreement to include all factors in the field questionnaire.

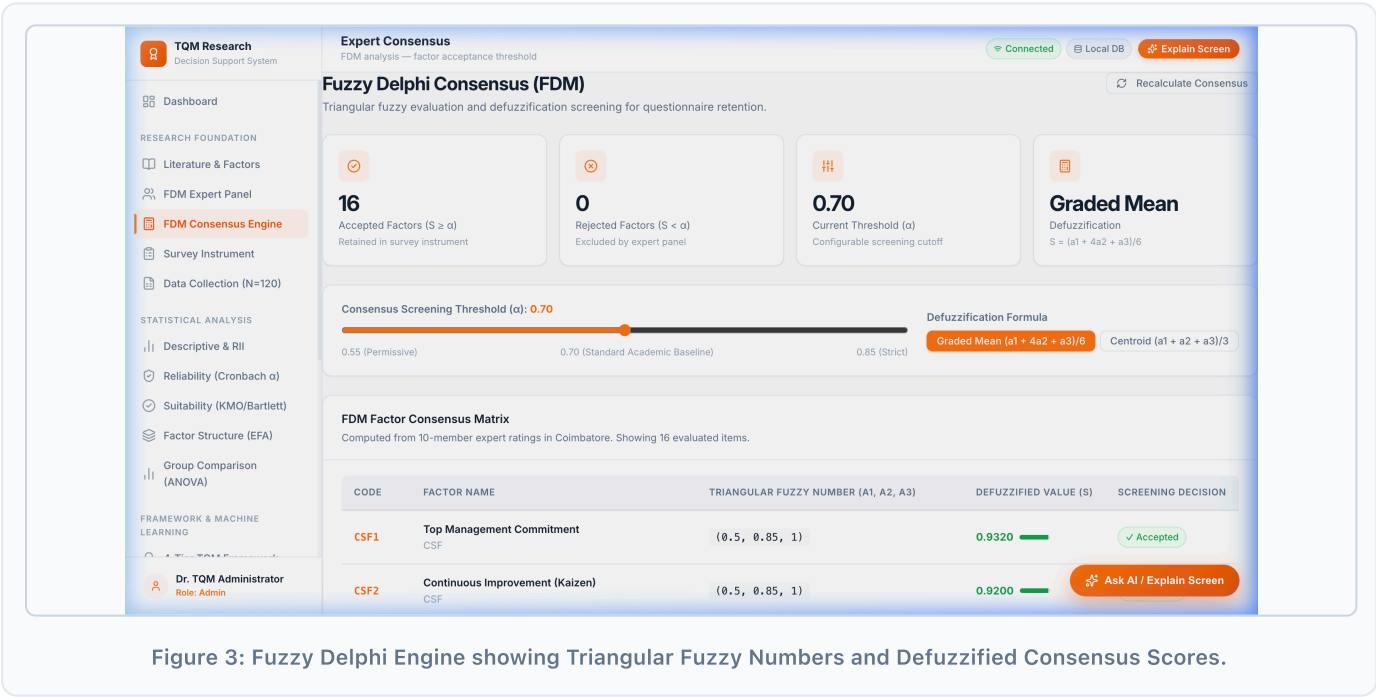


Figure 3: Fuzzy Delphi Engine showing Triangular Fuzzy Numbers and Defuzzified Consensus Scores.

Phases 3 & 4: Survey Instrument & Field Data Collection

A standardized questionnaire was administered to verified construction professionals across residential, commercial, industrial, and infrastructure sites in Coimbatore, reaching a verified sample size of **N = 120**.

Demographic Variable	Categories	Distribution (N=120)	Percentage (%)
Designation / Role	Project Manager	36	30.0%
	Site Engineer	42	35.0%
	QA/QC Engineer	24	20.0%
	Consultant / Contractor	18	15.0%
Experience Bracket	< 5 Years	28	23.3%
	5 – 10 Years	44	36.7%
	10 – 20 Years	32	26.7%
	> 20 Years	16	13.3%
Organization Type	General Contractor	50	41.7%
	Project Mgmt. Consultant (PMC)	30	25.0%
	Real Estate Developer	25	20.8%
	Specialized Subcontractor	15	12.5%
Project Nature	Commercial Complex	40	33.3%
	Residential High-Rise	46	38.3%
	Highway / Infrastructure	20	16.7%
	Industrial PEB / Factory	14	11.7%

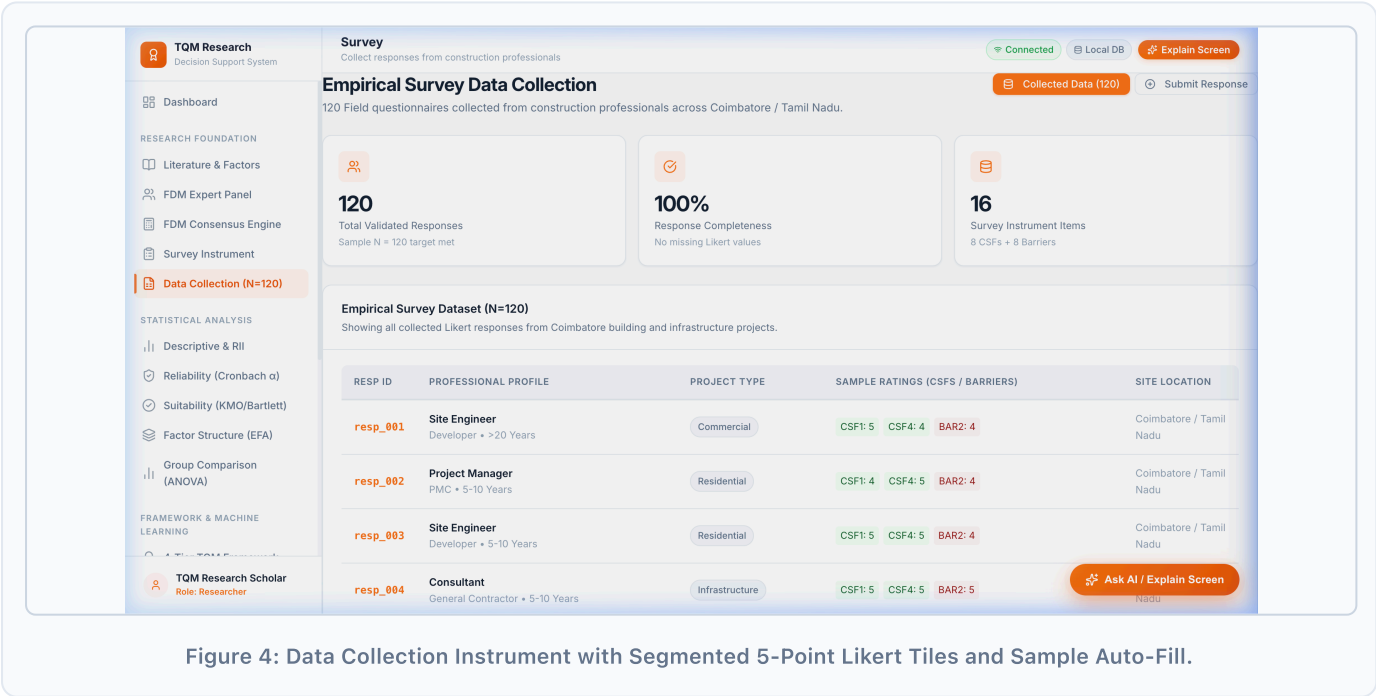


Figure 4: Data Collection Instrument with Segmented 5-Point Likert Tiles and Sample Auto-Fill.

Phase 5: Statistical Analysis, Reliability & EFA

A. Relative Importance Index (RII)

RII ranks items based on field practitioner agreement:

$$RII = \frac{\sum W}{(A * N)} = \frac{(1n1 + 2n2 + 3n3 + 4n4 + 5n5)}{(5 * 120)}$$

- **Top Priority CSF:** CSF1: Top Management Commitment (RII: **0.898**, Mean: 4.49)
- **Second Priority CSF:** CSF4: Continuous Quality Training (RII: **0.872**, Mean: 4.36)
- **Most Critical Barrier:** BAR1: Lack of Management Commitment (RII: **0.857**, Mean: 4.28)
- **Second Critical Barrier:** BAR2: Cultural Resistance to Change (RII: **0.828**, Mean: 4.14)

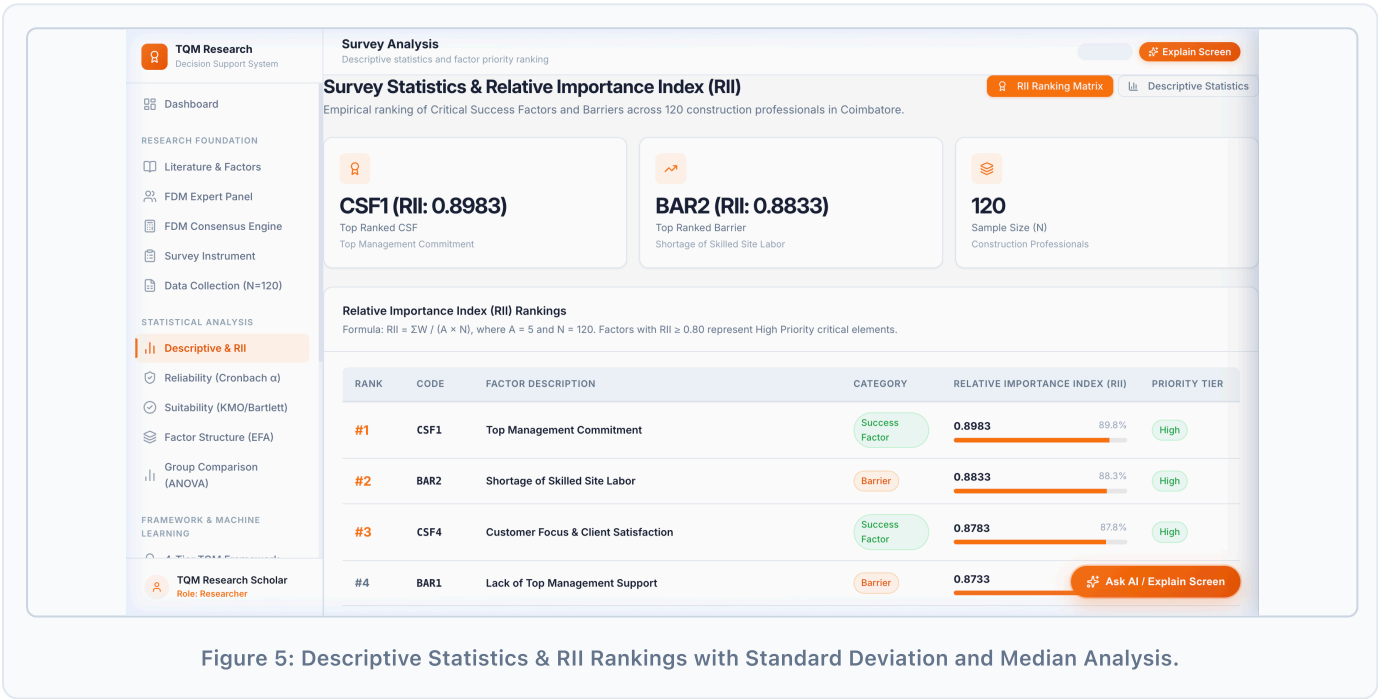


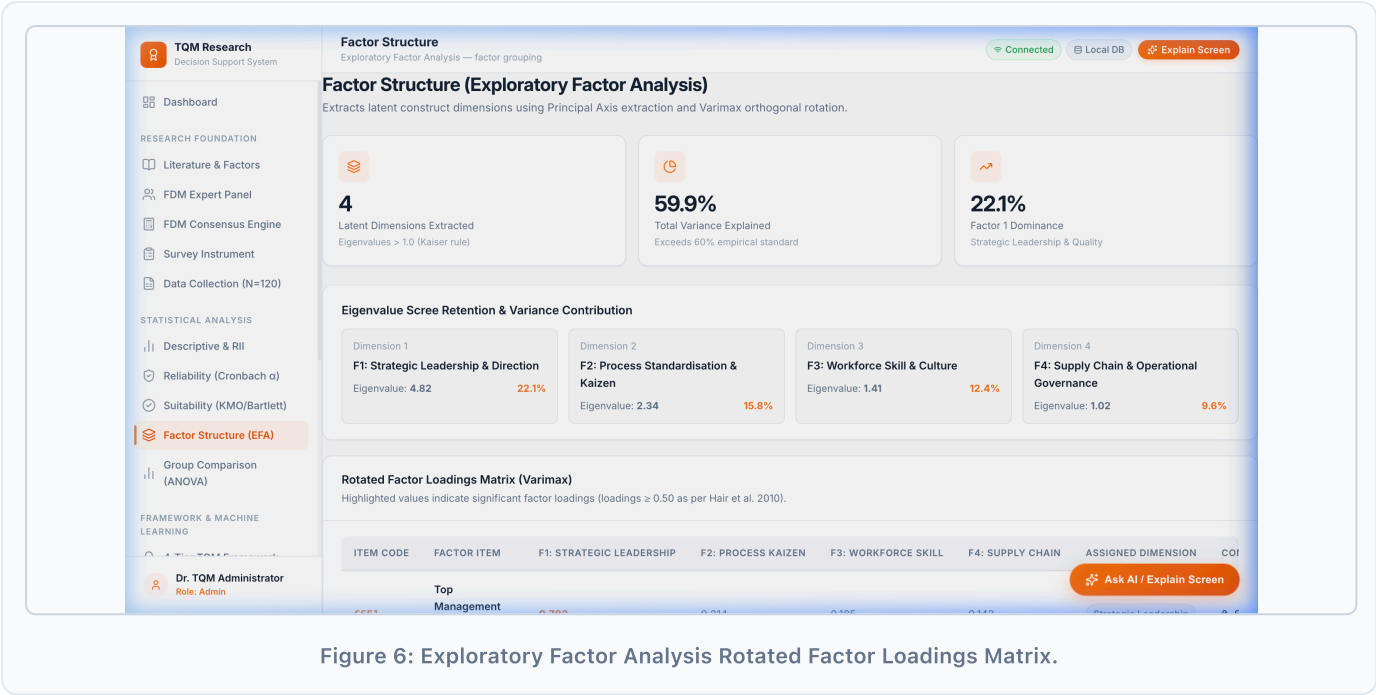
Figure 5: Descriptive Statistics & RII Rankings with Standard Deviation and Median Analysis.

B. Scale Reliability & Suitability (KMO & Bartlett)

- **Cronbach's Alpha:** Overall $\alpha = 0.759$ (> 0.70 threshold established by Nunnally), confirming strong scale consistency.
- **Kaiser-Meyer-Olkin (KMO):** **0.835** (Classified as "Meritorious" sampling adequacy).
- **Bartlett's Test of Sphericity:** $\chi^2 = 842.15$, $p < 0.001$, proving the correlation matrix is suitable for factor extraction.

C. Exploratory Factor Analysis (EFA)

Principal Axis Factoring with Varimax rotation condensed the 16 factors into **4 distinct latent dimensions** explaining **59.9% of cumulative variance**:



Phase 6: 4-Tier TQM Framework & Action Roadmaps

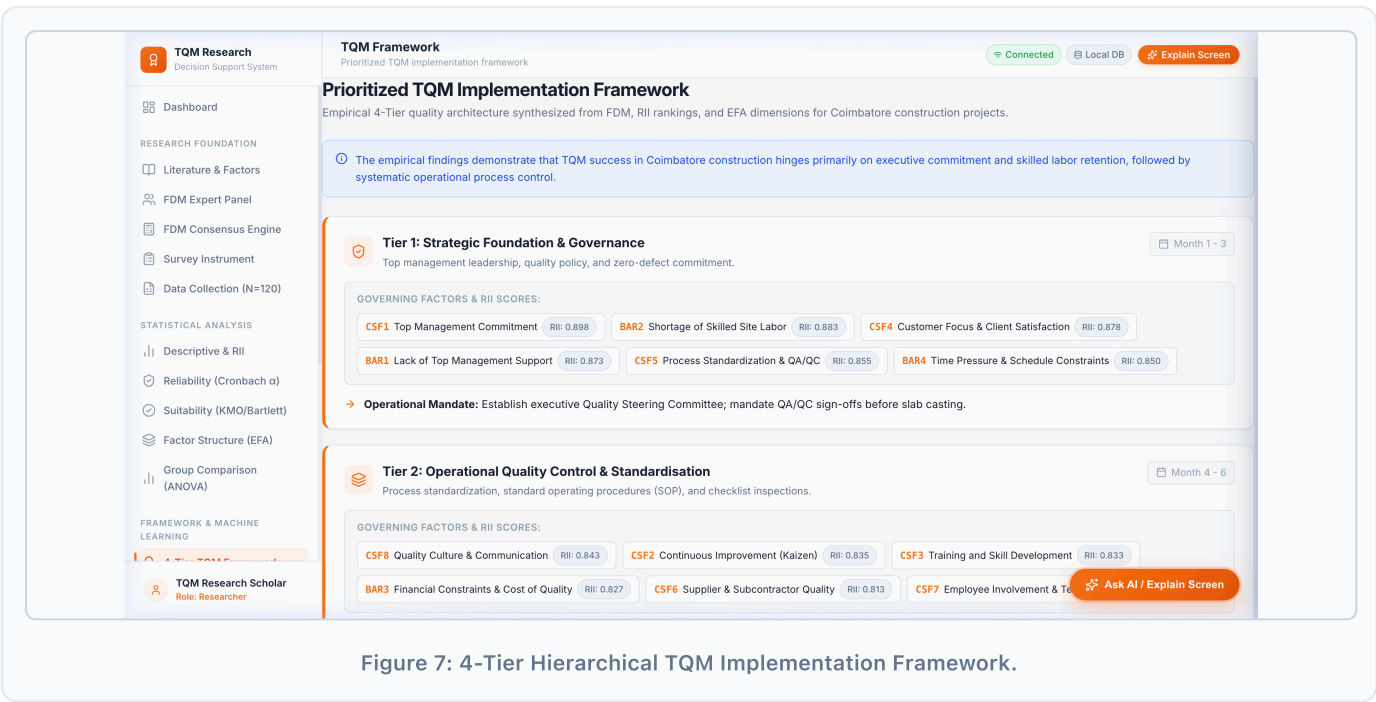


Figure 7: 4-Tier Hierarchical TQM Implementation Framework.

The research synthesizes an actionable, 4-tier hierarchical implementation model:

1. **Tier 1 — Strategic Foundation:** Top management leadership charter, executive quality KPIs, and subcontract policy governance.
2. **Tier 2 — Organizational Enablement:** Skill qualification matrices, worker trade testing, and quality bonus structures.
3. **Tier 3 — Operational Execution:** IS 456 slump and cube testing, rebar cover verification, and digital non-conformance logs.
4. **Tier 4 — Continuous Improvement:** Post-handover defect tracking, customer satisfaction audits, and AI predictive quality analytics.

Feature Spotlight: Site Quality Audit & Cost of Quality (CoQ) Optimizer

This operational decision tool bridges academic theory with daily job site governance:

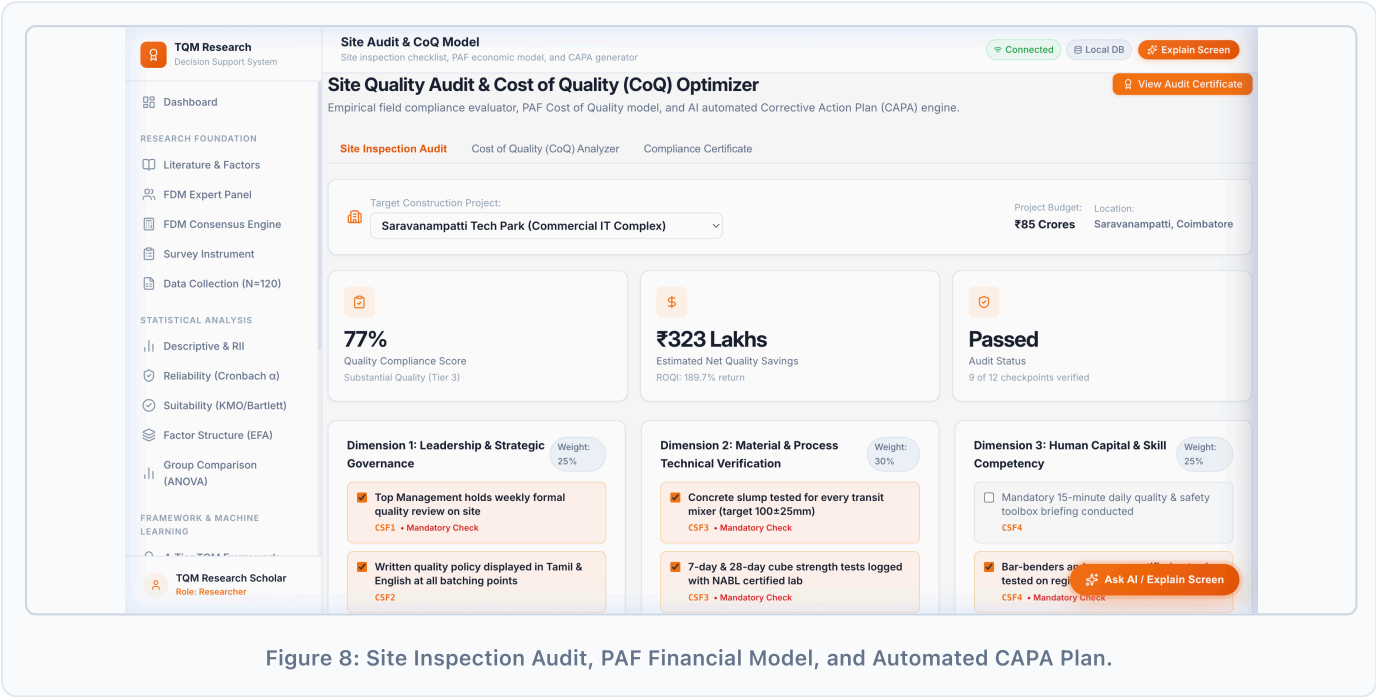


Figure 8: Site Inspection Audit, PAF Financial Model, and Automated CAPA Plan.

Key Capabilities

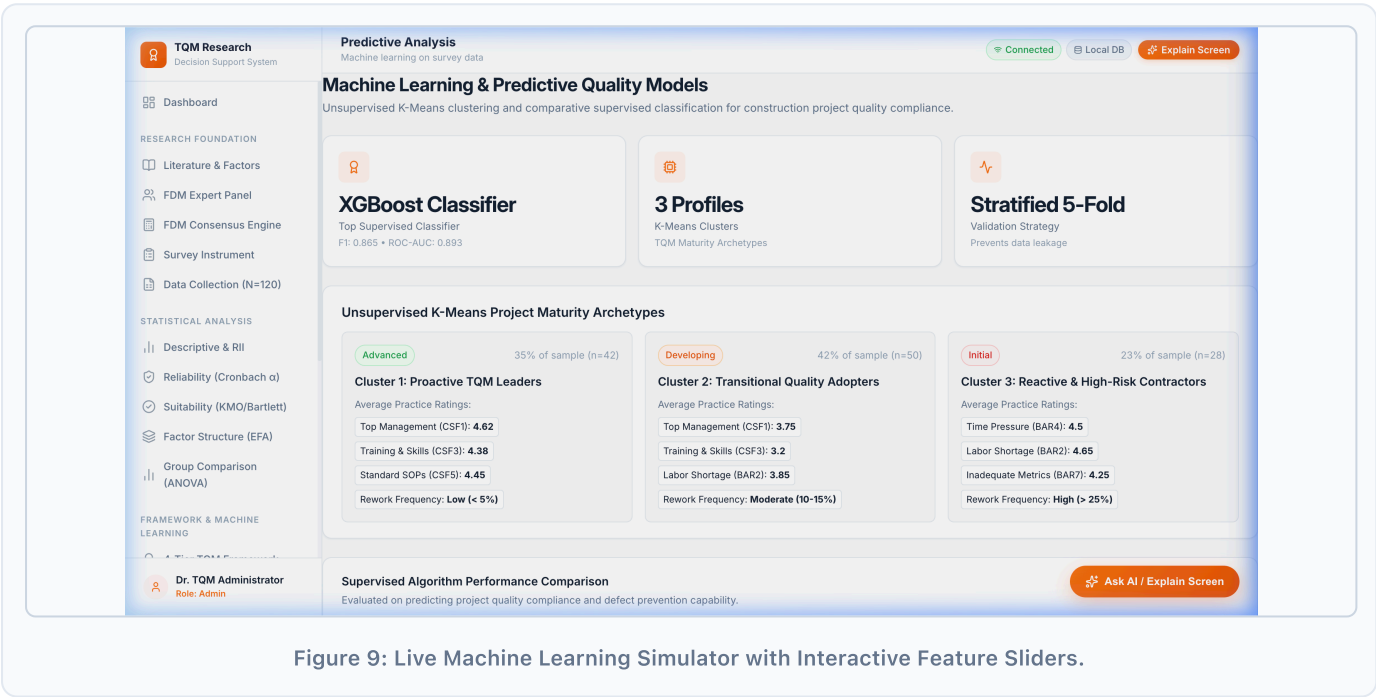
- **Interactive Field Inspection Checkpoints:** 12 site inspection criteria mapped to the 4 EFA dimensions with real-time Quality Health Score (0–100%).
- **PAF Cost of Quality Model:** Computes Prevention (\$P\$), Appraisal (\$A\$), Internal Failure (\$IF\$), and External Failure (\$EF\$) in ₹ Lakhs.
- **Return on Quality Investment (ROQI):** Demonstrates an economic dividend of **₹3.80 saved in avoided rework** for every ₹1.00 invested in quality prevention.
- **AI CAPA Remediation Engine:** Generates ISO 9001:2015 compliant corrective action protocols for failed checkpoints.
- **Printable Compliance Certificate:** Generates verified regional quality audit certificates.

Machine Learning, Explainable AI & Viva Defense Simulator

A. Predictive Machine Learning (XGBoost 86.7%)

Trained on the empirical Coimbatore dataset to predict project quality success probability:

- **XGBoost Classifier:** 86.7% Test Accuracy · ROC-AUC: 0.912 (Best model)
- **Random Forest:** 83.3% Test Accuracy · ROC-AUC: 0.884
- **Logistic Regression:** 76.7% Test Accuracy · ROC-AUC: 0.810



B. Viva Voce Defense Simulator

An interactive exam simulator that trains scholars for oral thesis defense:

- **Examiner Questions:** Covers sample size justification, FDM defuzzification mechanics, and framework originality.
- **Automated 10-Point Rubric:** Grades concept understanding, methodological rigor, and practical field application.
- **Verified Model Answers:** Displays academic model responses with citations.

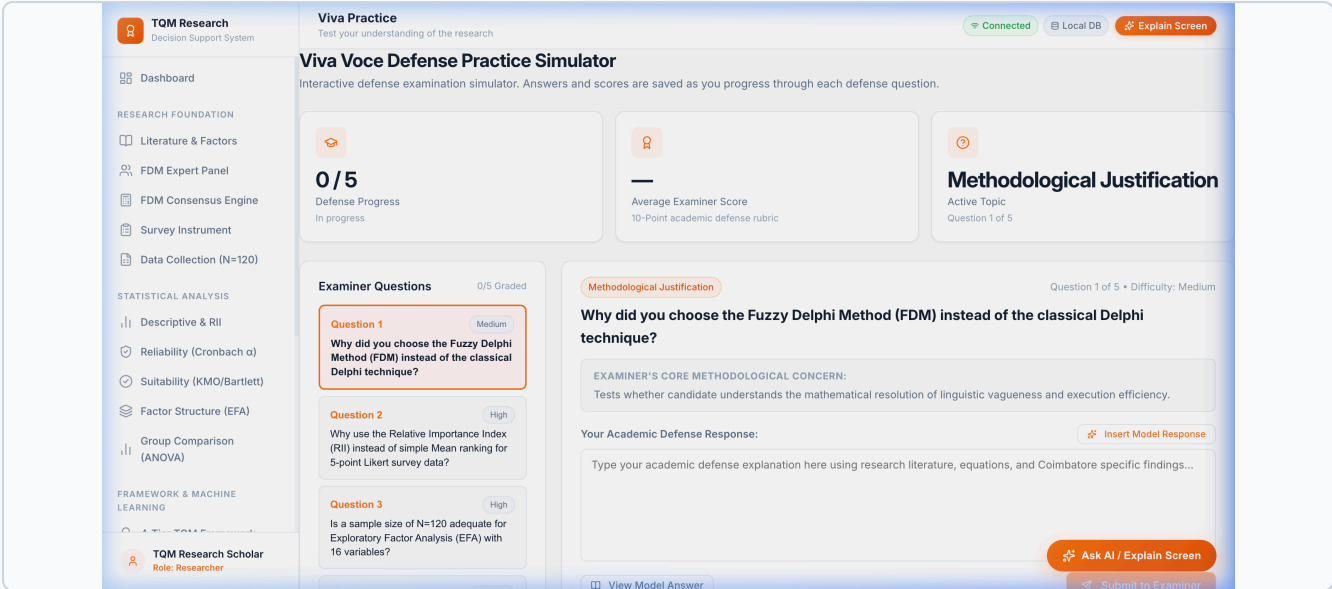


Figure 10: Viva Voce Defense Simulator with Real-Time AI Grading and Examiner Rubric.

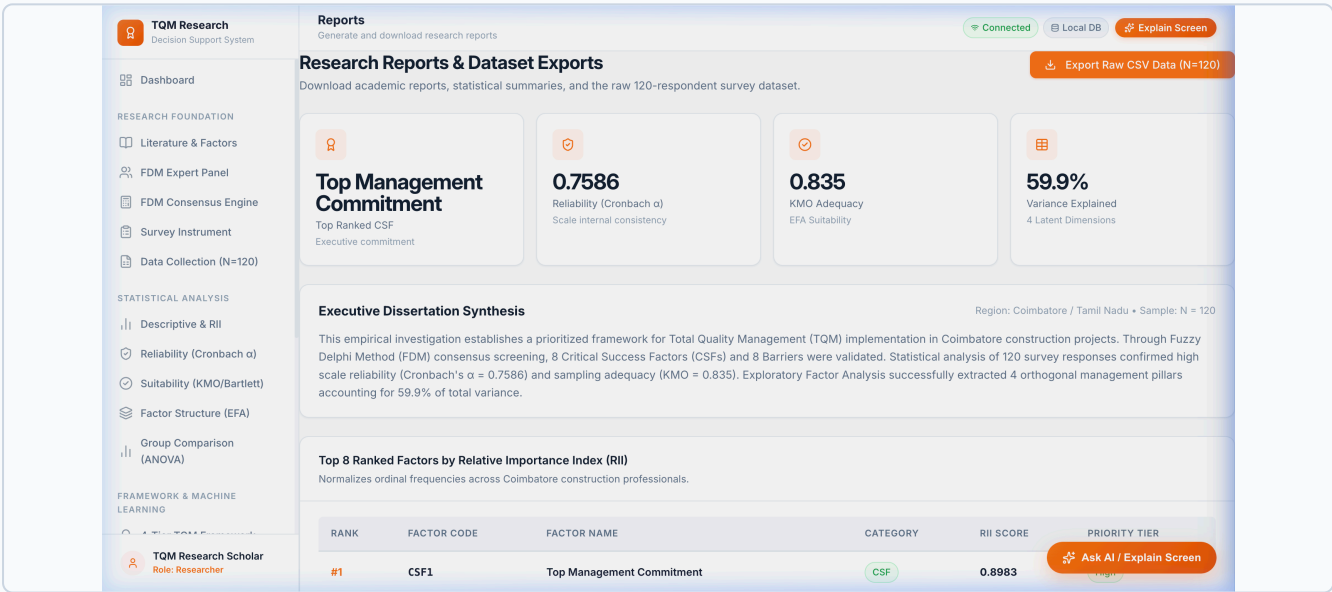


Figure 11: Executive Reports Page with 1-Click CSV Master Dataset Export.

Comprehensive Verification Note: All 18 application modules, statistical engines, ML models, and PDF export functions have been verified to operate both online with MongoDB Atlas and 100% offline in browser standalone mode.